

# Youssef Sawiris

Math Educator & Software Engineer

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github.com/ysawiris · linkedin.com/in/youssef-sawiris · YoYo Code Club (YouTube)

## SKILLS

### Languages

Python · TypeScript · JavaScript ·  
Scala · C++ · Go · Lua · Ruby ·  
SQL · Bash

### AI / Agents

Anthropic SDK · Claude Agent  
SDK · MCP servers · OpenAI ·  
Gemini · ElevenLabs · LLM tool-  
use

### Web

Next.js · React · Tailwind · Prisma ·  
Node.js · Django · Rails ·  
shadcn/ui

### Teaching

Project-based learning · IEP-  
aligned instruction · Small-group  
facilitation · AI literacy

## EDUCATION

### Reach University

M.A. Education + Teaching  
Credential (in progress)  
Expected 2027 · Oakland, CA

### Dominican University of California

B.S. Computer Science · Back-  
end concentration  
Jul 2021 · San Rafael, CA

## SELECTED PROJECTS

### Kordon

*Solo Builder*

Next.js e-commerce site for an  
aquarium/pond pet-products  
brand.

### FlowAgent

*Solo Builder*

AI workflow automation across  
Instagram, WhatsApp, Gmail, and  
Calendar.

### YoYo Code Club

Beginner Python & JS lessons (7  
eps, 2020–2021).

## PROFILE

Software engineer with nearly three years at Strava and a B.S. in Computer Science, now teaching 6th grade math in Oakland. My pedagogy leans on structured whole-class formative checks every lesson and IEP-aligned differentiation for the students on IEPs in my classes. I write production Python and ship AI-agent products on the side, and I love turning computer science into hands-on projects—especially for students who don't see themselves in tech yet.

## EXPERIENCE

### Lodestar Public Charter School · 6th Grade Math Teacher

*Jan 2025 – Present · Oakland, CA*

- Teach 6th grade math to ~25 students per period, building computational thinking and problem-solving aligned to CA Common Core standards.
- Use project-based, small-group facilitation with constant checks for understanding; differentiate and write IEP-aligned adaptations for the students on IEPs in my classes.
- Work primarily with first-generation, low-income students of color in urban Oakland—the same communities SMASH recruits from.
- Run structured whole-class formative checks every lesson—mini-whiteboards, half-sheet quick-checks, and IM cool-downs—to surface confusion in real time and adjust pacing.

### 2cm.ai · Software Engineering Consultant

*Feb 2024 – Present · Oakland, CA*

- Engaged on full-stack feature work for a multi-tenant Next.js / TypeScript SaaS for countertop fabricators.
- Recent engagements: canvas drawing tools, JSON pricing-rule schemas, Stripe Connect billing, and Three.js / React-PDF rendering for quotes and 3D visualizations.

### Strava · Software Engineer I & II

*Jun 2021 – Jan 2024 · San Francisco, CA*

- Re-indexed the Strava Search (“Surtch”) service to support 9B+ activities, improving query latency and unblocking new product features.
- Shipped the Best-Ever Efforts feature end-to-end, owning complex multi-system data backfills.
- Migrated the Garmin integration to a ping-and-pull pipeline processing 2M+ daily events.
- Led the Server Guild and mentored junior engineers on debugging, code review, and incident response.

### NVIDIA · Product Engineering Technician

*Dec 2015 – Aug 2016 · Santa Clara, CA*

- Identified GPU production quality issues on optical and electro-mechanical lines; partnered with engineering on root-cause analysis and component-level corrective actions.
- Assembled, aligned, and bench-tested pre-production graphic processors against SLT and ESD industry standards.